AI FAQ ASSISTANT - PROJECT REPORT

Phase 5: Project Development Phase

Project Setup

npm init -y
npm install express mongoose dotenv cors jsonwebtoken bcrypt @google/genai

Core Backend Modules

• server.js: Loads variables, connects DB, starts listening
• app.js: Creates Express app, enables CORS, mounts routes
• authMiddleware.js: JWT verification
• errorMiddleware.js: Centralized error handling

Database Schemas

User Schema:
- name (required)
- email (required, unique)
- password (hashed with bcrypt)
- role (default: 'user')
- timestamps

FAQ Schema:
- question (required)
- answer (required)
- category (enum validation)
- createdBy (user reference)
- timestamps

API Routes

Auth Routes:
• POST /api/auth/register (public)
• POST /api/auth/login (public)
• GET /api/auth/profile (protected)

FAQ Routes:
• GET /api/faqs (public)
• GET /api/faqs/search (public)
• GET /api/faqs/:id (public)

• POST /api/faqs (protected)
• PUT /api/faqs/:id (protected)
• DELETE /api/faqs/:id (protected)

AI Routes:
• POST /api/ai/generate-faq (protected)
• POST /api/ai/answer (protected)

Implementation Features

• Secure password hashing
• JWT token generation and verification
• Author-based FAQ modification
• Keyword-based search
• AI-powered content generation
• Request validation
• Error handling and logging